

Numerical Analysis — Homework 2

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Deadline: April 10, 2025

1. Consider the linear equation system $Ax = b$ with

$$A = \begin{pmatrix} -9.8 & 19.8 \\ -9.9 & 19.9 \end{pmatrix} \quad \text{and} \quad b = \begin{pmatrix} 10 \\ 10 \end{pmatrix}.$$

In the following, you may use that the inverse of the matrix A is given by

$$A^{-1} = \begin{pmatrix} 19.9 & -19.8 \\ 9.9 & -9.8 \end{pmatrix}.$$

- (a) What is the solution x of $Ax = b$?
- (b) Let $x + \delta x$ denote the solution of the perturbed equation $(A + \delta A)(x + \delta x) = b$ for $\|\delta A\|_\infty \leq 0.01$. Show that the relative error is bounded by

$$\frac{\|\delta x\|_\infty}{\|x\|_\infty} \leq 1.$$

2. Consider the symmetric and positive definite tri-diagonal matrix

$$A = \begin{pmatrix} a_1 & b_1 & & & \\ b_1 & \ddots & \ddots & & \\ & \ddots & \ddots & b_{n-1} & \\ & & b_{n-1} & a_n & \end{pmatrix} \in \mathbb{R}^{n \times n}.$$

- (a) Sketch a computer code that computes a Cholesky factorization, $A = LDL^\top$. Make sure that the computational complexity of your code is at most $O(n)$.
 - (b) What is the computational complexity of computing a QR-factorization of the above tri-diagonal matrix? Would you recommend using such a factorization?
3. Let the vectors $a_1 \in \mathbb{R}^2$ and $a_2 \in \mathbb{R}^2$ be given by

$$a_1 = \begin{pmatrix} 4 \\ -3 \end{pmatrix} \quad \text{and} \quad a_2 = \begin{pmatrix} 17 \\ 6 \end{pmatrix}.$$

We consider the standard scalar product in $\mathbb{R}^2$, $\langle x, y \rangle = x_1 y_1 + x_2 y_2$ for all $x, y \in \mathbb{R}^2$.

- (a) Explain how to use the Gram-Schmidt algorithm to find an orthonormal basis of the vector space $V = \text{span}(a_1, a_2)$.
- (b) Let $A = (a_1 \ a_2) \in \mathbb{R}^2$ be a matrix with columns a_1 and a_2 . Find an orthogonal matrix $Q \in \mathbb{R}^{2 \times 2}$ and an upper triangular matrix $R \in \mathbb{R}^{2 \times 2}$ such that $A = QR$.
- (c) Also explain how to use Householder's method for computing a QR-factorization of the matrix A .